AP1

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Overview

This project studies two vector norms (ℓ_1 and ℓ_∞) and their induced matrix norms. Two random 4-vectors were generated, reshaped into 2×2 matrices, and distances computed. Unit balls ($\|x - x_r\| \leq 1$) were visualized using 2D slices to illustrate the geometric behavior of each norm.

Norms Used

Vector Norms ($\mathbb{R}^4$)

Chosen for ease of visualization and manual implementation: $\ell_1 : \|x\|_1 = \sum |x_i|, \quad \ell_\infty : \|x\|_\infty = \max |x_i|$

Matrix Induced Norms (2×2)

Induced from vector norms: $\|A\|_1 =$ max column sum, $\|A\|_\infty =$ max row sum. All calculations were done manually without built-in functions.

Implementation

Vectors v_1, v_2 and matrices A_1, A_2 were generated with a fixed seed. Distances formulas:

$$d_v^{(1)} = \|v_1 - v_2\|_1, \quad d_v^{(\infty)} = \|v_1 - v_2\|_\infty$$

$$d_A^{(1)} = \|A_1 - A_2\|_1, \quad d_A^{(\infty)} = \|A_1 - A_2\|_\infty$$

Custom plotting functions made the code readable. Diamond shapes appear for ℓ_1, square shapes for ℓ_∞.

Observations & Insights

- ℓ_1 emphasizes total deviation, ℓ_∞ emphasizes the largest deviation.
- 2D slices help visualization but hide full 4D/matrix structure.
- Fixed seed ensures reproducible results; otherwise plots differ each run.
- Methods are general and work for any input vectors/matrices.

Conclusion

The project shows differences between vector and matrix norms and their geometric behavior. Manual computation strengthened understanding of distances, induced norms, and unit ball visualizations, while keeping the code flexible and reproducible.